

AI AGENT OPERATING SYSTEM

THE SYSTEM IS THE PRODUCT

USER INTENT

objective · context · constraints



CONTROL PLANE

route · plan · approve · evaluate



EXECUTION PLANE

tools · memory · state · outputs

From prompt chains to
controlled autonomy.

A field guide to choosing the
smallest architecture that can carry the work.

Autonomy is not a mood. It is a system boundary.

The design question is not “how agentic?” — it is “where should control live?”

WORK**What must be done**

task · context · definition of done

CONTROL**Who decides what happens next**

route · plan · permissions · escalation

LEARNING**How the system gets better**

trace · eval · feedback · release gate

A system earns autonomy only when its boundaries, feedback, and failure modes are visible.

Choose architecture by problem shape, not by fashion.

Two questions predict most of the design: can the steps be known, and can the work split safely?

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SINGLE AGENT

open-ended · tool use

ORCHESTRATED

specialists · synthesis

WORKFLOW

known steps · low variance

PARALLEL

independent branches

ANTI-PATTERN

**More agents
does not
mean
more system.**

Complexity is the cost.

PARALLELISM →

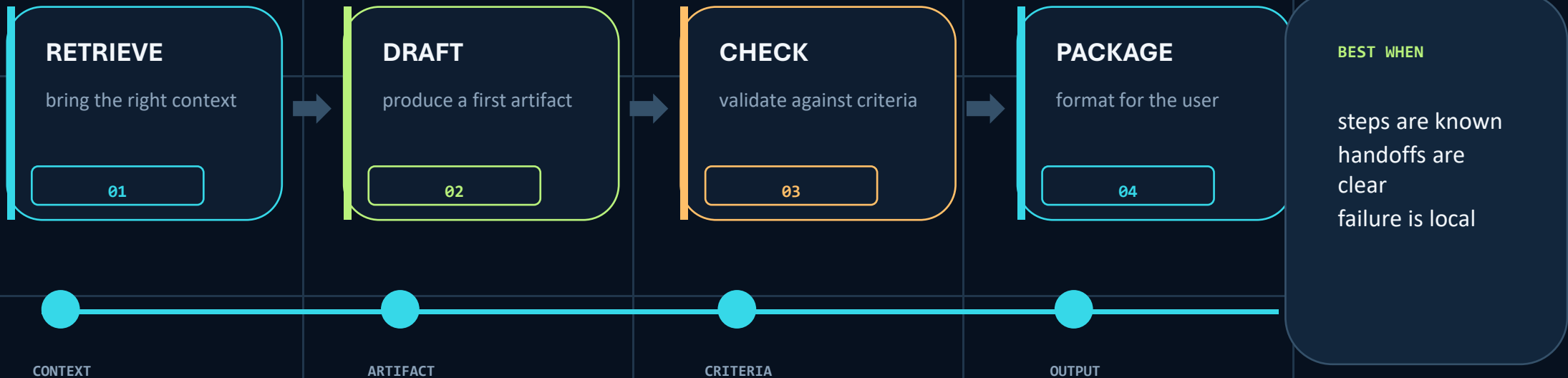
The operating system has two planes — and one feedback spine.

Control decides. Execution acts. Observability closes the loop.



Prompt chaining: make the path explicit.

Use a sequence when the work is stable enough to describe before execution.



Routing: send the task to the right lane.

A classifier can reduce complexity when the task family is recognizable.

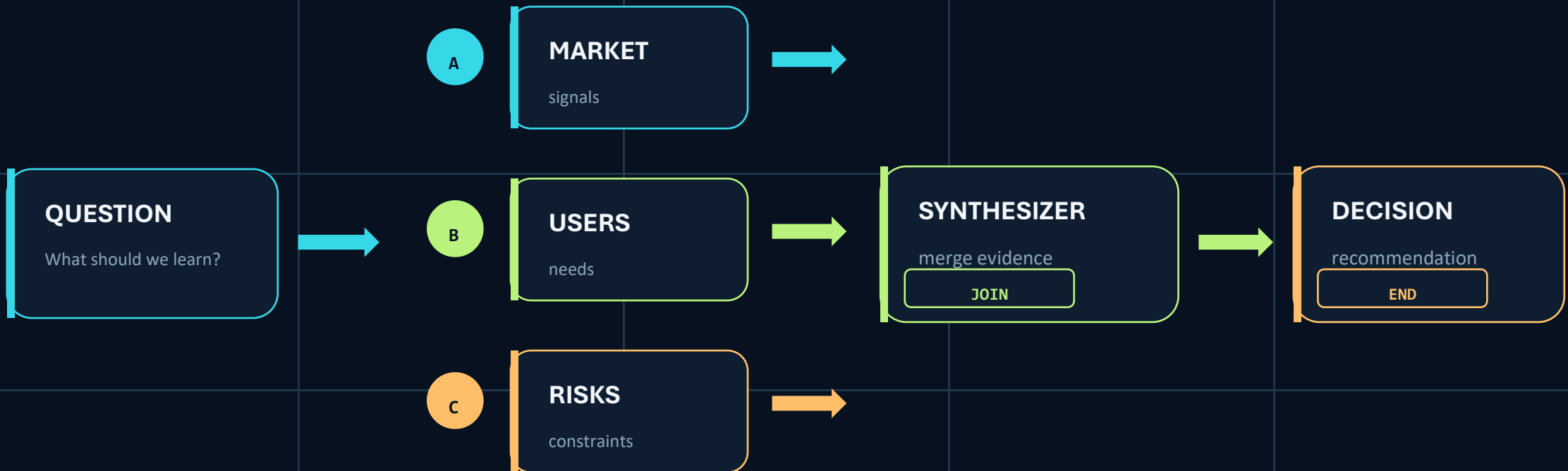


ROUTING IS A DECISION

The classifier should expose why a lane was chosen — not hide a second agent inside a black box.

Parallelization: split only where independence is real.

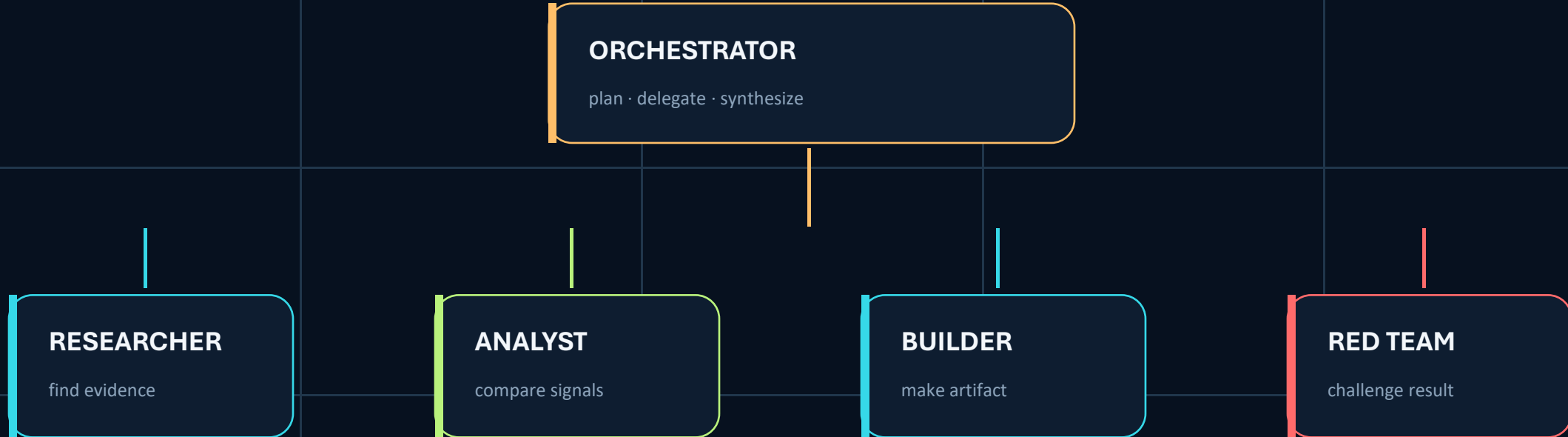
Fan out the work, then make synthesis an explicit step.



FAILURE MODE / parallel workers without a synthesis contract produce a pile of answers, not a system.

Orchestrator-workers: delegate the search space.

Use hierarchy when the task needs specialists and the plan must adapt as evidence arrives.

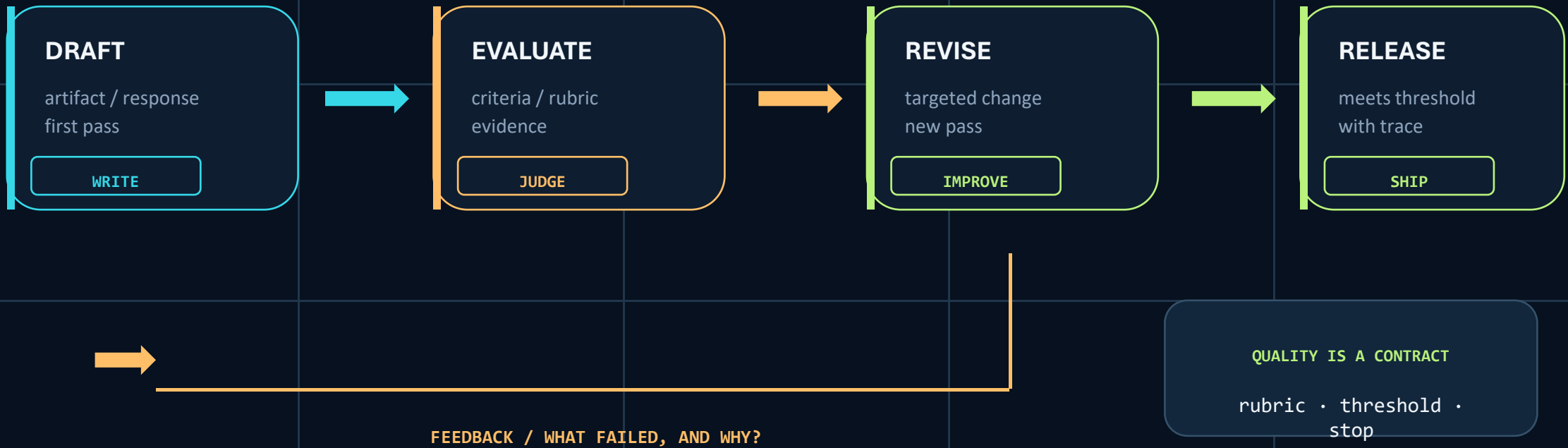


The orchestrator owns the objective. Specialists own the search space.

ADAPTIVE PLAN / SHARED CONTEXT

Evaluator-optimizer: make quality a loop.

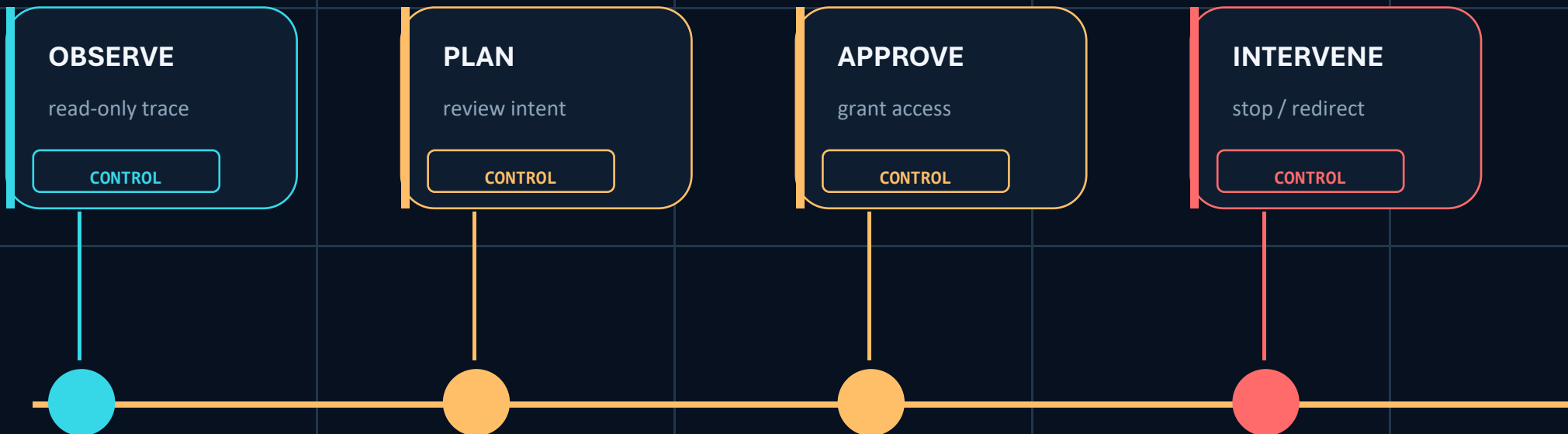
A useful agent does not only produce an answer; it knows what a better answer means.



Human control belongs at the level of intent.

The safest review point is usually the plan, the permission boundary, and the final release — not every tool call.

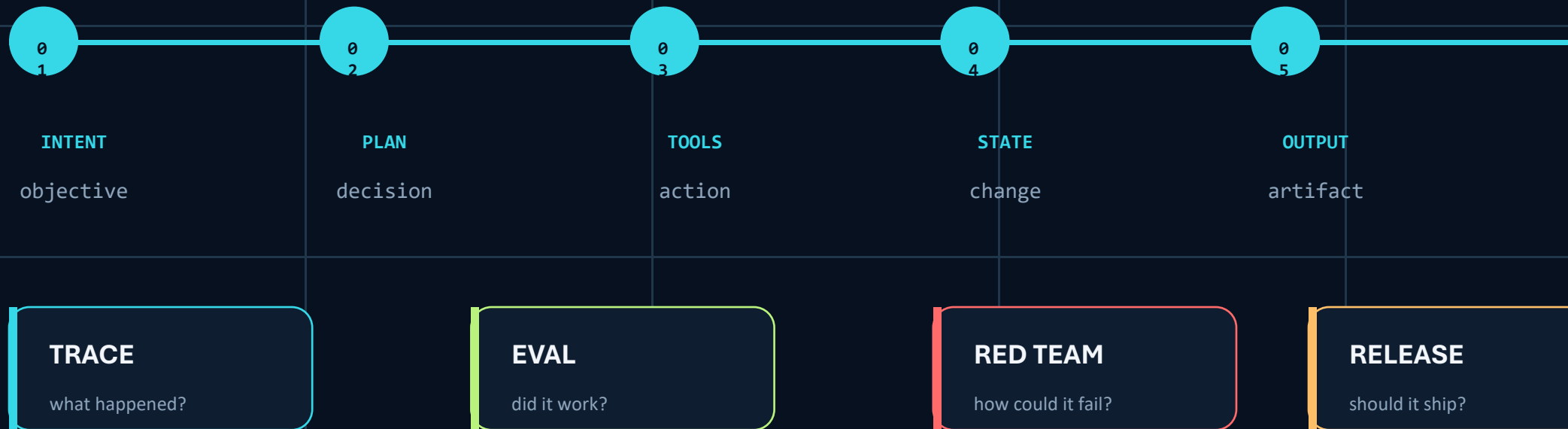
A good control surface compresses many low-level actions into a small number of high-value decisions.



If you cannot trace it, you cannot improve it.

Agent quality lives across turns, tools, state changes, and final artifacts.

OBSERVABILITY IS NOT LOGGING. IT IS THE EVIDENCE LAYER FOR SYSTEM BEHAVIOR.



Build the smallest system that can carry the work.

Complexity is not the proof of intelligence. Fit is.

Start small. Earn autonomy.

Choose the path.
Expose the boundary.
Close the loop.

IF THE WORK IS...

START WITH...

predictable

workflow

branching

router

parallel

fan-out

open-ended

agent loop

high-stakes

human gate

THE OPERATING SYSTEM IS THE DECISION SURFACE BETWEEN INTENT AND ACTION.